

Cansu Düzgün

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Education

Ph.D., Meteorology	Florida State University	Expected 2026
	Dissertation: “Evaluation of Environmental Parameters that Influence Convective Entrainment Rates” Committee: Henry E. Fuelberg (Advisor), Allison Wing, Christopher Holmes, Guosheng Liu, Rebecca Adams-Selin	
M.S., Meteorology	Florida State University	2023
	Thesis: “Quantifying Simulated Deep Convective Transport and Its Sensitivity to Lightning Data Assimilation in a Supercell and Mesoscale Convective System” Advisor: Henry E. Fuelberg	
B.S., Meteorological Engineering	Istanbul Technical University	2021
	Thesis: “Comparison of Different Gridded-Precipitation Products with Observations Over Turkey and Extreme Precipitation Analysis” Advisor: Yurdanur S. Unal Engineering Project: “Design of Sediment Chamber for Control of Cement Factory Air Contaminants” Advisor: Ceyhan Kahya	

Professional Experience

Graduate Research Assistant	Florida State University, Tallahassee, FL	1/2022 – Present
Meteorological Engineer	Buluttan WI, Istanbul, Turkey	9/2021 – 1/2022

Publications

- 3 R. Adams-Selin, **C. Düzgün**, N. Heath, H. E. Fuelberg, J. Hegarty (In prep). Comparison of convective vertical transport sensitivity to parameterization characteristics.
- 2 **Düzgün, C.**, H. E. Fuelberg, R. Adams-Selin, N. Heath, 2026: Evaluating Parameterized Vertical Transport in Varying Deep Convective Modes: A Comparative Analysis of Supercell vs MCS. *Mon. Wea. Rev.*, <https://doi.org/10.1175/MWR-D-25-0113.1>.
- 1 **Düzgün, C.**, H. E. Fuelberg, R. Adams-Selin, N. Heath, 2025: Evaluating the Effectiveness of Lightning Data Assimilation in Parameterized Deep Convection. *Mon. Wea. Rev.*, <https://doi.org/10.1175/mwr-d-24-0221.1>.

Conference Presentations

- 7 **Düzgün, C.**, H. E. Fuelberg, R. Adams-Selin, N. Heath, Simulating Assessing Subgrid Convective Transport and Mass Flux in Deep Convective Clouds Using Parameterized WRF-Chem Simulations, 105th AMS Annual Meeting, New Orleans, LA, January 12-16th 2025. [oral]
- 6 **Düzgün, C.**, H. E. Fuelberg, R. Adams-Selin, N. Heath*, Evaluating the Use of Lightning Data Assimilation in WRF to Improve the Simulation of Parameterized Deep Convection, Joint WRF/MPAS Users Workshop 2024, Boulder, CO, June 25-27th 2024. [oral]
*Presenting author.
- 5 **Düzgün, C.**, H. E. Fuelberg, R. Adams-Selin, N. Heath, Simulating Troposphere-Stratosphere Exchange and Chemical Transport during Supercell and Mesoscale Convective Events, 104th AMS Annual Meeting, Baltimore, MD, January 28 – February 1st 2024. [oral]

- 4 **Düzgün, C.**, H. E. Fuelberg, R. Adams-Selin, N. Heath, Simulating Deep Convective Transport and Its Sensitivity to Lightning Data Assimilation, AGU23 Annual Meeting, San Francisco, CA, December 11-15th 2023. [oral, invited]
- 3 **Düzgün, C.**, H. E. Fuelberg, R. Adams-Selin, N. Heath, J. Hegarty, Simulations of Deep Convective Transport to the UTLS, 103rd AMS Annual Meeting, Denver, CO, January 8-12th 2023. [poster]
- 2 **Düzgün, C.**, C. Y. Sonuc, Y. S. Unal, Comparison of Different Gridded-Precipitation Products with Observations over Turkey, International Global Climate Change Congress (IGCCC), Canakkale, Turkey, March 3-5th 2021. [oral, virtual conference]
- 1 **Düzgün, C.**, B. Ozturk, E. Tan, O. Sen, Air Quality Impact By Aircrafts in LTO number Record-Breaking Year 2016 at Istanbul Atatürk Airport by using the WRF-Chem Model, 7th International Meeting on Meteorology and Climatology of the Mediterranean, Palma de Mallorca, Spain, March 4-6th 2019. [poster]

Co-authored Presentations

- 6 Adams-Selin, R., N. Heath, **C. Düzgün**, H. E. Fuelberg, J. Hegarty, Quantification of the Importance of Parameterized Deep Convective Characteristics to Successful Simulation of Convective Transport, 105th AMS Annual Meeting, New Orleans, LA, January 12-16th 2025. [oral]
- 5 J. Hegarty, Adams-Selin, R., **C. Düzgün**, N. Heath, H. E. Fuelberg, Simulations of Entrainment in Continental Thunderstorms: Implications for Parameterization of Deep Convective Vertical Transport, 104th AMS Annual Meeting, Baltimore, MD, January 28 – February 1st 2024. [oral]
- 4 Adams-Selin, R., J. Hegarty, N. Heath, **C. Düzgün**, H. E. Fuelberg, Exploration of Parameterized and Explicitly Simulated Convective Properties for a DC3 Case Study, 103rd AMS Annual Meeting, Denver, CO, January 8-12th 2023. [oral]
- 3 Kursun, I., S. Guneslik, **C. Düzgün**, R. Ozturk, I. Kelkitli, Y. Urkmez, N. Ekinici, Sensitivity of WRF Parameterizations on Lightning Events in Istanbul, 8th International Meeting on Meteorology and Climatology of the Mediterranean, Virtual, May 25-27th 2021. [oral]
- 2 Deniz, G., **C. Düzgün**, E. Tan, Saharan Dust Transport Modeling for Turkey by using the WRF-Chem Model, 7th International Meeting on Meteorology and Climatology of the Mediterranean, Palma de Mallorca, Spain, March 4-6th 2019. [poster]
- 1 Dinc, U., B. Toker, Z. F. Unal, I. Kursun, **C. Düzgün**, G. Deniz, H. Toros, Air Pollution Prediction in Istanbul by using WRF-CHEM, WRF-Chem EU Workshop, Istanbul, Turkey, May 2-4th 2018. [oral]

Honors and Awards

Dean's Award for Doctoral Excellence (DADE)	2024, 2025
Chi Epsilon Pi Meteorology Honor Society Induction	2024 - Present

Professional Society Membership

American Meteorological Society	2022 - Present
American Geophysical Union	2023 - Present

Outreach

Capital Regional Science and Engineering Fair Judge

Judge for middle and high school student projects presented in Earth and Environmental Science in 2024, 2025.

Skills

WRF, WRF-Chem, Python, Fortran90, NCL, Bash, Git/GitHub, MET/METplus, ArcGIS, Linux/Unix, HPC systems (SLURM), Conda, Microsoft Office, Google Suite, scientific writing, leadership, presentation.

Languages

Turkish (Native), English (Fluent), Italian (Beginner)